

Gradient, divergence, and rotation in different coordinate systems

In Cartesian Coordinate System: $dl : dx, dy, dz$

vector : $\vec{A}$ or $\mathbf{A}$

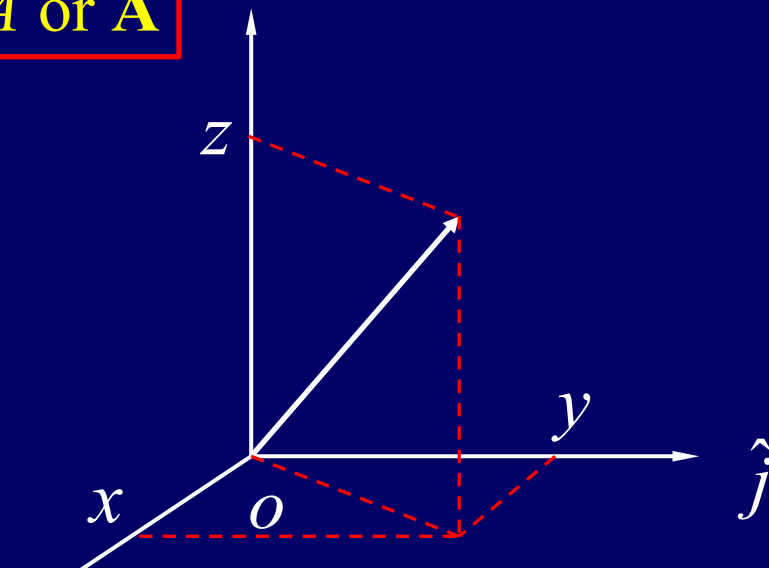
$ds : dxdy, dydz, dxdz...$

$dV : dxdydz$

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

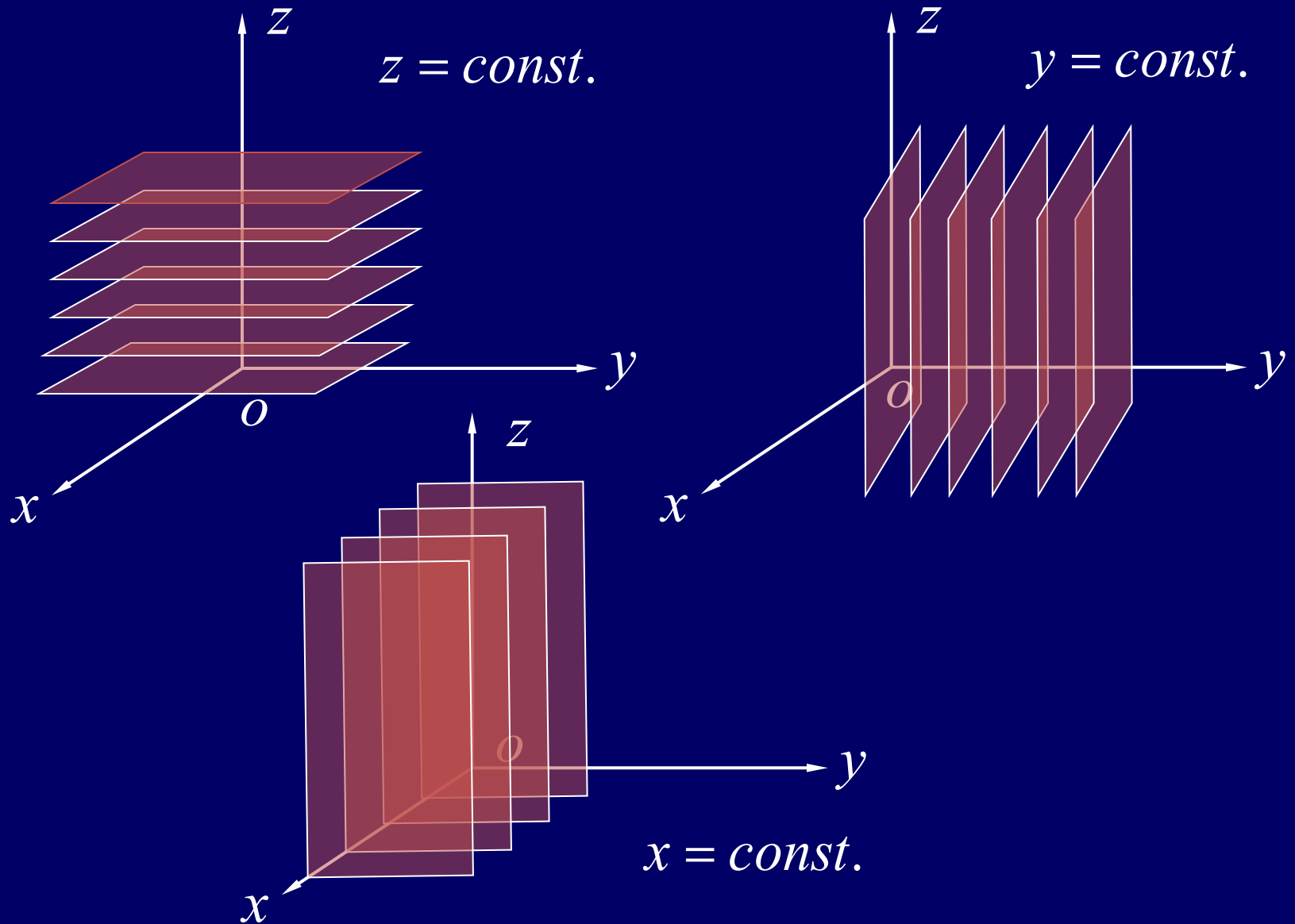
$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$\begin{aligned} \nabla \times \mathbf{A} = & \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{j} \\ & + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{k} \end{aligned}$$



$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

In Cartesian Coordinate System

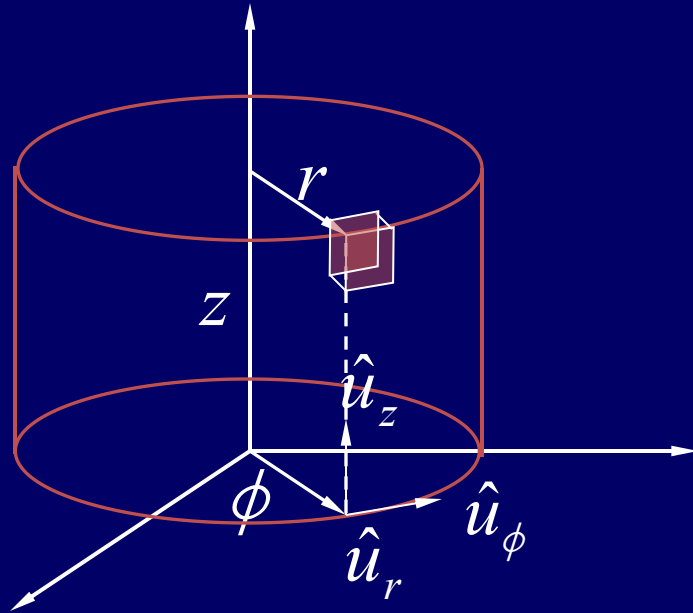


In Cylindrical Coordinate System

$$dl : dr, r d\phi, dz$$

$$ds : r d\phi dz$$

$$dV : r dr d\phi dz$$



$$s = \int ds = \int_0^L \int_0^{2\pi} r d\phi dz = 2\pi r L$$

$$V = \int dV = \int_0^R \int_0^L \int_0^{2\pi} r dr d\phi dz = \pi R^2 L$$

$$\hat{u}_r \frac{\partial}{\partial r}, \quad \hat{u}_z \frac{\partial}{\partial z}, \quad \hat{u}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} \quad \nabla = \hat{u}_r \frac{\partial}{\partial r} + \hat{u}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{u}_z \frac{\partial}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

$$\nabla \cdot \mathbf{A} = (\hat{u}_r \frac{\partial}{\partial r} + \hat{u}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \hat{u}_z \frac{\partial}{\partial z}) \cdot (A_r \hat{u}_r + A_\phi \hat{u}_\phi + A_z \hat{u}_z)$$

$$\nabla \cdot \mathbf{A} = (\hat{u}_r \frac{\partial}{\partial r} + \hat{u}_\phi \frac{1}{r} \frac{\partial}{\partial \phi}) \cdot (A_r \hat{u}_r + A_\phi \hat{u}_\phi) + \frac{\partial A_z}{\partial z}$$

$$\begin{aligned} \nabla \cdot \mathbf{A} = & \hat{u}_r \frac{\partial A_r}{\partial r} \cdot \hat{u}_r + \cancel{\hat{u}_r \cdot A_r \frac{\partial \hat{u}_r}{\partial r}} + \cancel{\hat{u}_r \frac{\partial A_\phi}{\partial r} \cdot \hat{u}_\phi} + \cancel{\hat{u}_r \cdot A_\phi \frac{\partial \hat{u}_\phi}{\partial r}} \\ & + \cancel{\hat{u}_\phi \frac{1}{r} \frac{\partial A_r}{\partial \phi} \cdot \hat{u}_r} + \frac{A_r}{r} \hat{u}_\phi \cdot \frac{\partial \hat{u}_r}{\partial \phi} + \hat{u}_\phi \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} \cdot \hat{u}_\phi + \cancel{\hat{u}_\phi \cdot A_\phi \frac{\partial \hat{u}_\phi}{\partial \phi}} + \frac{\partial A_z}{\partial z} \end{aligned}$$

We have
already used

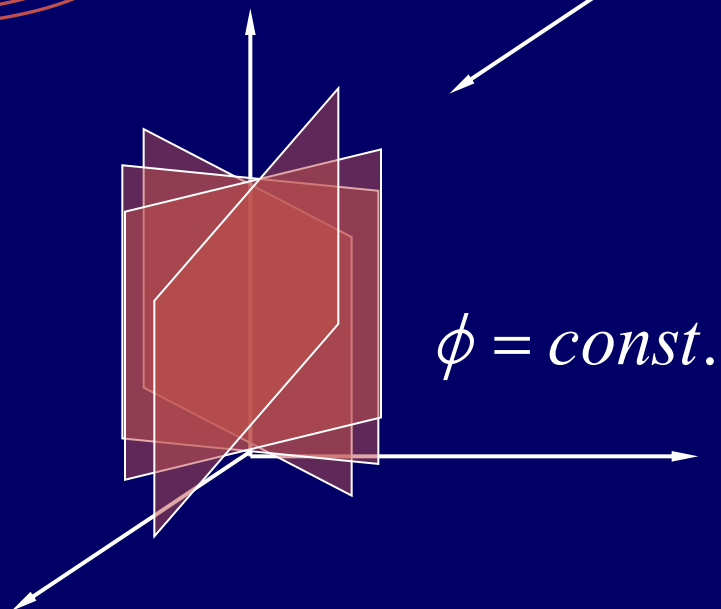
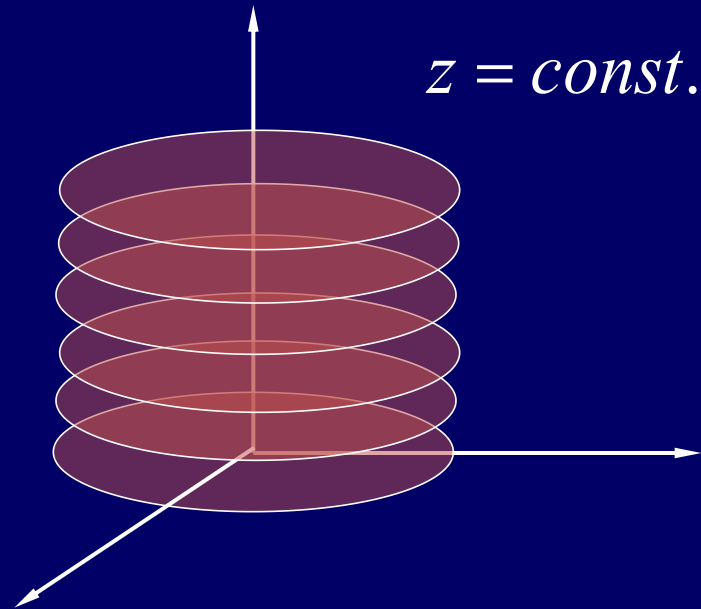
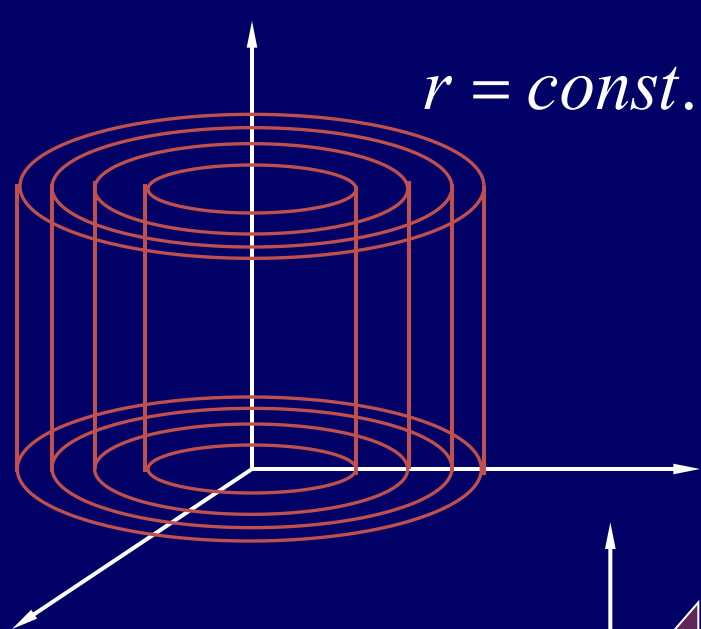
$$\hat{u}_r = \cos \phi \hat{i} + \sin \phi \hat{j} \longrightarrow \frac{\partial \hat{u}_r}{\partial \phi} = \hat{u}_\phi \quad \frac{\partial \hat{u}_r}{\partial r} = 0$$

$$\hat{u}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j} \longrightarrow \frac{\partial \hat{u}_\phi}{\partial \phi} = -\hat{u}_r \quad \frac{\partial \hat{u}_\phi}{\partial r} = 0$$

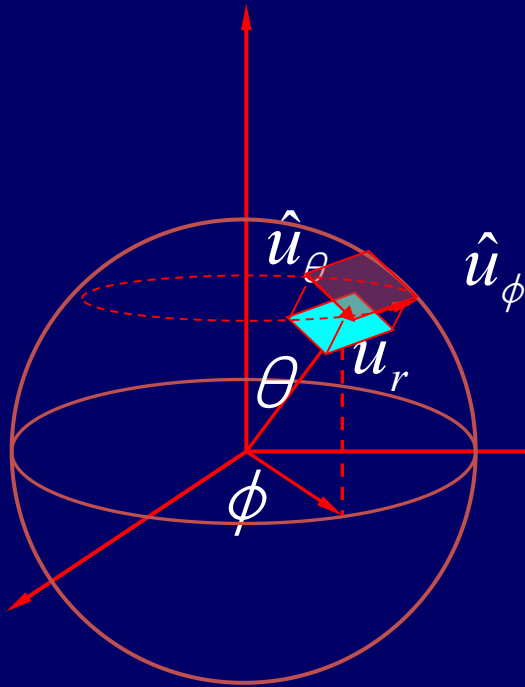
Therefore

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial}{\partial \phi} A_\phi + \frac{\partial}{\partial z} A_z$$

In Cylindrical Coordinate System



In Spherical Coordinate System



$$dl : dr, r d\theta, r \sin \theta d\phi$$

$$ds : r^2 \sin \theta d\theta d\phi$$

$$dV : r^2 \sin \theta d\theta d\phi dr$$

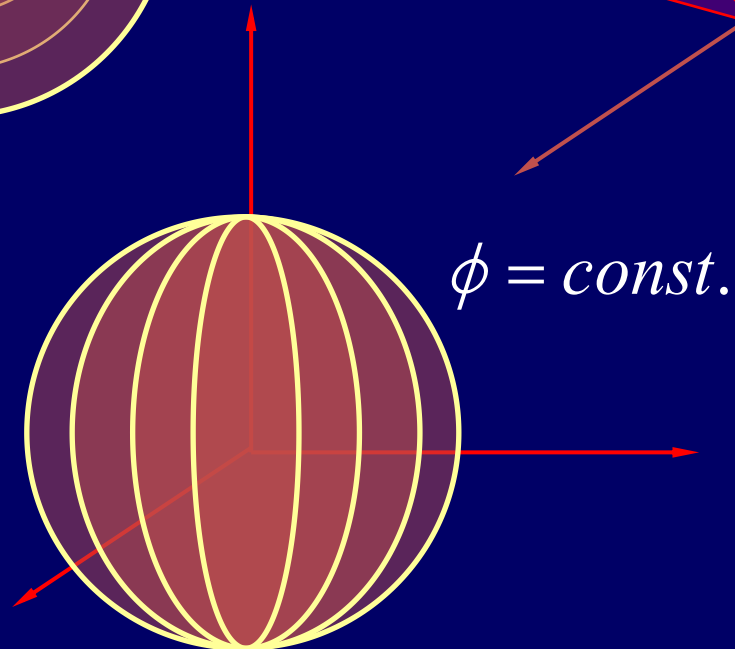
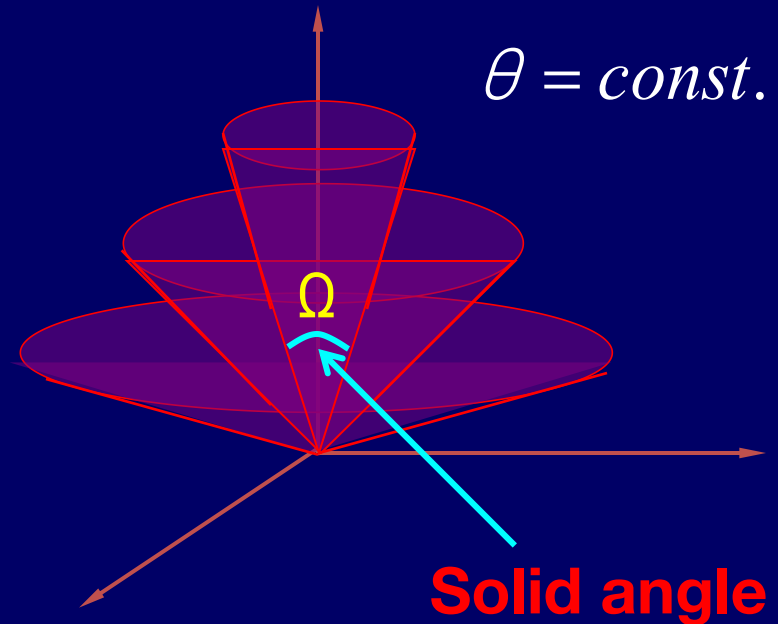
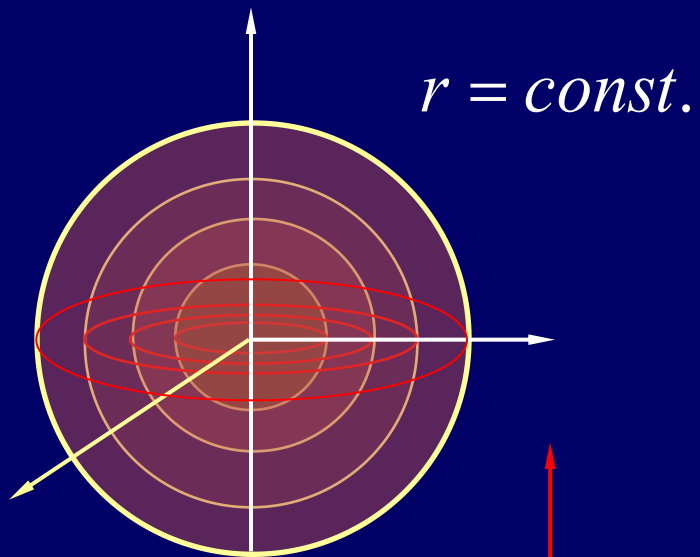
$$s = \int ds = \int_0^{2\pi} \int_0^\pi r^2 \sin \theta d\theta d\phi = 4\pi r^2$$

$$V = \int dV = \int_0^R \int_0^{2\pi} \int_0^\pi r^2 \sin \theta d\theta d\phi dr = \frac{4\pi R^3}{3}$$

$$\hat{u}_r \frac{\partial}{\partial r}, \quad \hat{u}_\theta \frac{1}{r} \frac{\partial}{\partial \theta}, \quad \hat{u}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \quad \nabla = \hat{u}_r \frac{\partial}{\partial r} + \hat{u}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{u}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} A_\phi$$

In Spherical Coordinate System



$$\Omega = \frac{A}{r^2}$$

In Cylindrical Coordinate System

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{u}_r & \hat{u}_\phi & \hat{u}_z \\ \frac{\partial}{\partial r} & \frac{1}{r} \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & A_\phi & A_z \end{vmatrix}$$
$$\hat{u}_r = \cos \phi \hat{i} + \sin \phi \hat{j}$$
$$\hat{u}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j}$$

In Spherical Coordinate System

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{u}_r & \hat{u}_\theta & \hat{u}_\phi \\ \frac{\partial}{\partial r} & \frac{1}{r} \frac{\partial}{\partial \theta} & \frac{\partial}{r \sin \theta \partial \phi} \\ A_r & A_\theta & A_\phi \end{vmatrix}$$
$$\hat{u}_r = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}$$
$$\hat{u}_\theta = -\cos \theta \cos \phi \hat{i} - \cos \theta \sin \phi \hat{j} + \sin \theta \hat{k}$$
$$\hat{u}_\phi = -\sin \phi \hat{i} + \cos \phi \hat{j}$$

Simple Tensor Analysis

Tensor: $\mathbf{T} = \vec{T} = T_{xx}\hat{i}\hat{i} + T_{xy}\hat{i}\hat{j} + T_{xz}\hat{i}\hat{k}$
 $+ T_{yx}\hat{j}\hat{i} + T_{yy}\hat{j}\hat{j} + T_{yz}\hat{j}\hat{k}$
 $+ T_{zx}\hat{k}\hat{i} + T_{zy}\hat{k}\hat{j} + T_{zz}\hat{k}\hat{k}$

Unit tensor: $\mathbf{I} = \vec{I} = \hat{i}\hat{i} + \hat{j}\hat{j} + \hat{k}\hat{k}$

Dyad(并矢): $\mathbf{AB} = A_x B_x \hat{i}\hat{i} + A_x B_y \hat{i}\hat{j} + A_x B_z \hat{i}\hat{k}$
 $+ A_y B_x \hat{j}\hat{i} + A_y B_y \hat{j}\hat{j} + A_y B_z \hat{j}\hat{k}$
 $+ A_z B_x \hat{k}\hat{i} + A_z B_y \hat{k}\hat{j} + A_z B_z \hat{k}\hat{k}$

$\mathbf{AB} \neq \mathbf{BA}$

$$(\mathbf{AB}) \cdot \mathbf{C} = \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$$

$$\mathbf{C} \cdot (\mathbf{AB}) = (\mathbf{C} \cdot \mathbf{A})\mathbf{B}$$

Thus $(\mathbf{AB}) \cdot \mathbf{C} \neq \mathbf{C} \cdot (\mathbf{AB})$

$$(\mathbf{AB}) : (\mathbf{CD}) = (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$$

$$\vec{\vec{I}} \cdot \mathbf{A} = \mathbf{A} \cdot \vec{\vec{I}} = \mathbf{A}$$

$$\nabla \cdot (\mathbf{AB}) = (\nabla \cdot \mathbf{A})\mathbf{B} + (\mathbf{A} \cdot \nabla)\mathbf{B}$$